



Patterns

(Class Slides)

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Nested Loops

Loop inside a loop

	c1	c2	c3	c4
Row 1	1	1	1	1
R2	2	2	2	2
R3	3	3	3	3
R4	4	4	4	4

$n = 4$

n

① Outer loop : no. of Rows

② Inner loop : no. of columns /
each row

③ Work in inner loop (each row)
i print

Print Star pattern

n = 4

R1 * 1st
 R2 ** 2st
 R3 *** 3st
 R4 **** 4st

① outer loop → Rows (i) → Row no.
 n times (1 to n)

② inner loop (each row's)
 columns
 i times (1 to i)

③ work?

cout << "*" << endl

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        cout << "*" << " ";
    }
    cout << endl;
}
```

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Print **Inverted Star** pattern

$n = 4$
 $4 = n - i + 1$
 $3 = n - i + 1$
 $2 = n - i + 1$
 $1 = n - i + 1$

R1 * * * *
 R2 * * *
 R3 * *
 R4 *

- ① outer loop (rows)
(1 to n)
- ② inner loop (each row)
(1 to $n - i + 1$)
- ③ work?
cout << "*" << endl

Print Half Pyramid pattern

$n=4$

R1	1	1	
R2	12	1 to 2	<u>1 to i</u>
R3	123	1 to 3	
R4	1234	1 to 4	

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j << " ";
    }
    cout << endl;
}
```

① outer loop (rows)
(1 to n)

② inner loop (each row)
(1 to i) $\Rightarrow j$

③ work?

$cout << j ;$

Print Character Pyramid pattern

A
BC
DEF
GHIJ

$n = 4$

char ch = 'A'

① outer loop (row)
(1 to n)

② inner loop (each row)
(1 to i)

③ work?

cout << ch;

ch++;
}

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Print Hollow Rectangle pattern

<u>R1</u>	* * * * *	1st + 3st + 1st
R2	* _ _ _ *	1st + 3sp + 1st
R3	* _ _ _ *	1st + 3sp + 1st
<u>R4</u>	* * * * *	1st + 3st + 1st

first or last $\rightarrow$ 5 stars
(1) (n)

① Outer loop (rows)
(1 to n)

② Inner loop (each row)

cout << "*" ; // First

for (1 to n-1) {
1st or last $\rightarrow$ "*" } ② work
else $\rightarrow$ " "

}
cout << "*" ; // last

Inverted & Rotated Half-Pyramid

$n=4$

— — — *	R1	3 sp	+ 1 st
— — **	R2	2 sp	+ 2 st
— ***	R3	1 sp	+ 3 st
****	R4	0 sp	+ 4 st

$i=1$

$n-i$ spaces
 $4-1=3$

① outer loop (rows)
 (1 to n)

② inner loop (each row)

a) Spaces (1 to $n-i$)
`cout << " " ← work`

b) Stars (1 to i)
`cout << "*" ← work`

`cout << endl`

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Print Floyd's Triangle

R1 1
 R2 2 3
 R3 4 5 6
 R4 7 8 9 10
 R5 11 12 13 14 15

1 el
 2 el
 3 el
 4 el
 5 el

ith i times
 (1 to i)

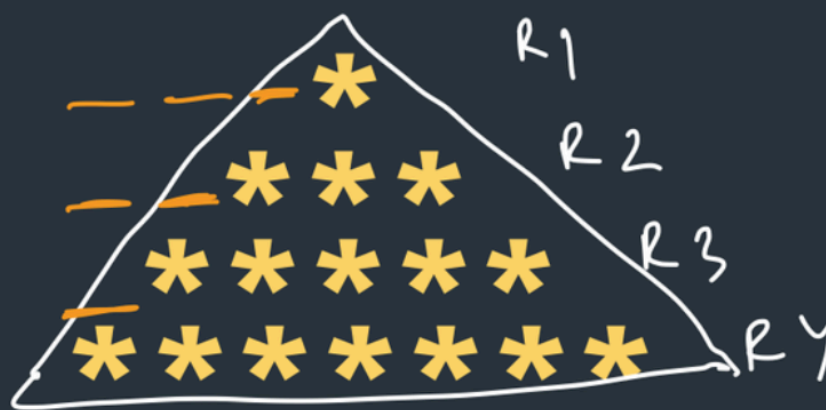
num = 1

- ① outer loop (rows)
(1 to n)
- ② inner loop (each row)
elements
(1 to i)
- ③ work?
cout << num;
num++;

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Diamond Pattern

$n=4$



R_1 3sp + 1st
 R_2 2sp + 3st
 R_3 1sp + 5st
 R_4 0sp + 7st

1st pyramid

① outer loop (rows)
(1 to n)

② inner loop (each row elements)

a) (1 to $n-i$) cout << " "
 b) (1 to $2*i-1$) cout << "*"

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Diamond Pattern

$n=4$

* * * * *
 - * * * *
 - - * * *
 - - - *
 R₄
 R₃
 R₂
 R₁

0 sp + 7 st
 1 sp + 5 st
 2 sp + 3 st
 3 sp + 1 st

2nd pyramid

① outer loop (rows)
(n to 1)

② inner loops (each row elements)

a) sp (1 to $n-i$)

b) st (1 to $2 \times i - 1$)

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Print Butterfly Pattern

$n=4$

```

  *      *
 **     **
 ***    ***
 ****   ****
 *****
 ****   ****
 ***    ***
 **     **
 *      *
  
```

```

  *      *
 **     **
 ***    ***
 ****   ****
 *****
 ****   ****
 ***    ***
 **     **
 *      *
  
```

R1 : 1st + 6sp + 1st
 R2 : 2st + 4sp + 2st
 R3 : 3st + 2sp + 3st
 R4 : 4st + 0sp + 4st

↓ ↓ ↓

Pattern1

① outer loop (rows)
(1 to n)

② inner loop

a) stars (1 to i)

b) spaces (1 to $2 \times (n-i)$)

c) stars (1 to i)

Print Butterfly Pattern

$n=4$

```

  *      *
 **     **
 ***    ***
 ****   ****
 *****

```

```

  *      *
 **     **
 ***    ***
 ****   ****
 *****

```

```

*****
****  **
***   **
**    **
*     *

```

R4 4 st
R3 3 st
R2 2 st
R1 1 st

0 sp 4 st
2 sp 3 st
4 sp 2 st
6 sp 1 st

Pattern

① outer loop (rows)
(n to 1)

② inner loop

a) stars (1 to i)

b) spaces (1 to $2 \times (n-i)$)

c) stars (1 to i)

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n = 4

$$2 * (n - i) = 2 * (4 - 1) = 6$$

i = 1



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